# 3.6 Carboxylic Acids & Derivatives

## Question Paper

| | |
|---|---|
| Course | CIE A Level Chemistry |
| Section | 3. Organic Chemistry |
| Topic | 3.6 Carboxylic Acids & Derivatives |
| Difficulty | Medium |

**Time allowed:** 20

**Score:** /10

**Percentage:** /100

## Question 1

Compound **P** is the **only** organic material required to produce the ester $CH_3CH_2CO_2CH_2CH_2CH_3$ through a sequence of reactions.

What might be the identity of compound **P**?

1 $CH_3COCH_3$

2 $CH_3CH_2CHO$

3 $CH_3CH_2CH_2OH$

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

**[1 mark]**

## Question 2

Sour milk contains a compound called lactic acid (2–hydroxypropanoic acid) which has the molecular formula $CH_3CH(OH)CO_2H$.

Which of the following reactions could lactic acid undergo?

A. $CH_3CH(OH)CO_2H + Cl_2 \rightarrow CH_3CH(Cl)CO_2H + OCl$

B. $CH_3CH(OH)CO_2H + HCO_2H \rightarrow CH_3CH(O_2CH)CO_2H + H_2O$

C. $CH_3CH(OH)CO_2H + CH_3OH \rightarrow CH_3CH(OCH_3)CO_2H + H_2O$

D. $CH_3CH(OH)CO_2H + NaHCO_3 \rightarrow CH_3CH(ONa)CO_2H + H_2O + CO_2$

**[1 mark]**

## Question 3

The organic compound, **C**, reacts with calcium metal to produce a salt with the empirical formula $CaC_2H_3O_2$.

What could be the identity of **C**?

1 Ethanoic acid

2 Butanedioic acid

3 2–methylpropanedioic acid

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

**[1 mark]**

## Question 4

Cetyl palmitate is a waxy solid and has the molecular formula $C_{15}H_{31}CO_2C_{16}H_{33}$.

Which products are formed when cetyl palmitate is heated under reflux with an excess of aqueous sodium hydroxide?

A. $C_{15}H_{31}CO_2Na$ and $C_{16}H_{33}OH$

B. $C_{15}H_{31}CO_2Na$ and $C_{16}H_{33}ONa$

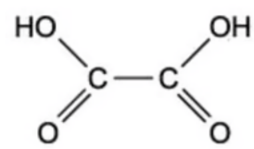
C. $C_{15}H_{31}OH$ and $C_{16}H_{33}CO_2Na$

D. $C_{15}H_{31}ONa$ and $C_{16}H_{33}CO_2Na$

**[1 mark]**

## Question 5

The diagram shows the structure of ethanedioic acid.



Ethanedioic acid reacts with ethanol in the presence of a few drops of concentrated sulfuric acid to form a diester. The molecular formula of the diester is $C_6H_{10}O_4$.

What is the structural formula of the diester?

A. $CH_3CH_2CO_2CO_2CH_2CH_3$

B. $CH_3CH_2OCOCO_2CH_2CH_3$

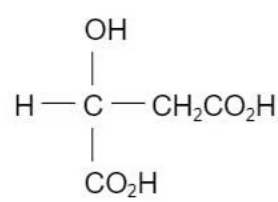
C. $CH_3CH_2O_2CO_2CCH_2CH_3$

D. $CH_3CO_2CH_2CH_2OCOCH_3$

**[1 mark]**

## Question 6

Apples contain a compound called malic acid.



malic acid

Which substance will react with only one hydroxyl groups in malic acid under suitable conditions?

A. Na (s)

B. $PCl_5$

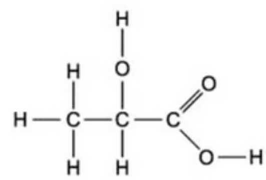
C. NaOH (aq)

D. $Cr_2O_7^{2-}$ / $H^+$ (aq)

**[1 mark]**

## Question 7

Lactic acid is a naturally occurring molecule.




Lactic acid

What is a property of lactic acid?

A. It rapidly decolourises aqueous bromine.

B. Two molecules can react together in strong acid.

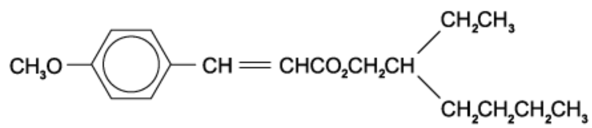
C. It reduces Fehling's reagent.

D. It is insoluble in water.

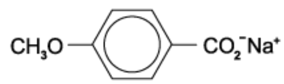
**[1 mark]**

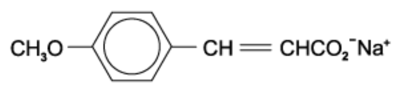
## Question 8

The ester in the diagram below is the active ingredient in sun protection creams.



The ester is hydrolysed by aqueous sodium hydroxide. Which substances are present in the products?

1 

2 

3 $CH_3CH_2CH_2CH_2CH(CH_2CH_3)CH_2OH$

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

**[1 mark]**

## Question 9

Compound **Q** has the formula $CH_3CH_2CO_2CH_3$.

What is the name of compound **Q** and how does its boiling point compare with that of butanoic acid?

| | Name of Q | Boiling point compared to butanoic acid |
|---|---|---|
| **A** | Methyl propanoate | Lower |
| **B** | Propyl methanoate | Lower |
| **C** | Methyl propanoate | Higher |
| **D** | Propyl methanoate | Higher |

**[1 mark]**

## Question 10

How many ester compounds have the molecular formula $C_4H_8O_2$?

A. 2

B. 3

C. 4

D. 5

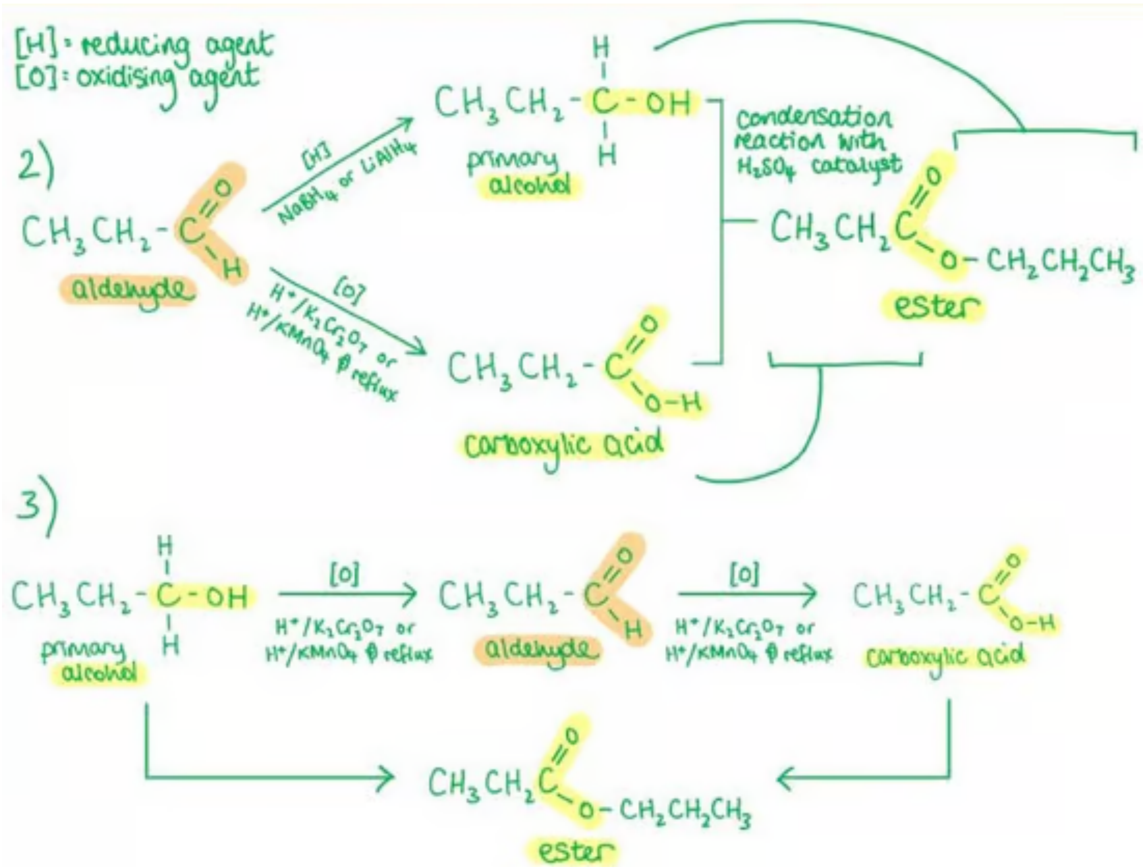
**[1 mark]**

# Model Answers: Medium

1

The correct answer is **C** because:

- **Ester** formation requires a **carboxylic acid** and an **alcohol.**
- For compound **P** to be the sole organic reactant to produce esters it must be capable of converting into an alcohol **and** a carboxylic acid.
- Compound 2 is the **aldehyde** propanal, which can be both reduced into a primary alcohol, propan-1–ol, and oxidised to form a carboxylic acid, propanoic acid.
- Compound 3 is the primary alcohol propan-1–ol which can be oxidised to the carboxylic acid propanoic acid.
- Propan-1–ol and propanoic acid will undergo a **condensation reaction** in the presence of a **sulfuric acid ($H_2SO_4$) catalyst** to produce an ester.

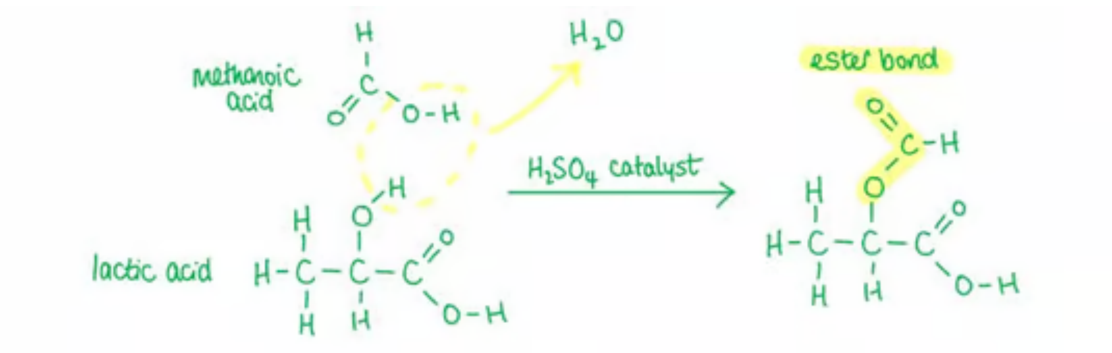


**A, B** & **D** are incorrect as all these options contain compound 1, propanone, which is a ketone and therefore will not oxidise to a carboxylic acid and therefore cannot produce esters.

2

The correct answer is **B** because:

- The lactic acid has a secondary alcohol group and a carboxylic acid group.
- Reacting lactic acid with **methanoic acid**, $HCO_2H$, will produce an **ester** from the secondary alcohol group as show in reaction B.
- This reaction is a condensation reaction and therefore produces water alongside the organic product.




**A** is incorrect as neither the carbon-oxygen bond in the secondary alcohol and the chlorine-chlorine bond in $Cl_2$ have enough of a **dipole** to enable chlorine to substitute into the compound, this reaction is more likely if **hydrogen chloride**, $HCl$, reacts with a **tertiary alcohol** to produce the haloalkane and $\mathbf{H_2O}$**.**

**C** is incorrect as methanol, $CH_3OH$, would react with the **carboxylic acid** in lactic acid to produce an ester $CH_3CH(OH)\mathbf{CO_2CH_3}$**.**

**D** is incorrect as sodium metal would **replace the hydrogen** in the **carboxylic acid** group to produce $CH_3CH(OH)\mathbf{CO_2^-Na^+}$**.**

3

The correct answer is **B** because:

- Carboxylic acids take part in **redox** reactions with **reactive metals** such as calcium and sodium to produce a **salt** and **hydrogen gas.**

- The molecular formula is the actual number of atoms of each element in a molecule.
- The empirical formula is the lowest ratio of atoms of each element in a molecule.
- When reacted with calcium:
  - Ethanoic acid, $CH_3COOH$, would convert to $CH_3COOCa$ which has the molecular and empirical formula $\mathbf{CaC_2H_3O_2}$**.**
  - Butanedioic acid, $HOOCCH_3CH_3COOH$, would convert to $CaOOCCH_3CH_3COOCa$ which has the molecular formula $Ca_2C_4H_6O_4$ and the empirical formula $CaC_2H_3O_2$.

**A** & **C** are incorrect as **2-methylpropanedioic acid**, $HOOCCH(CH_3)COOH$, would convert to $CaOOCCH(CH_3)COOCa$ which has the molecular formula $\mathbf{Ca_2C_4H_4O_4}$ and the empirical formula $\mathbf{CaC_2H_2O_2}$**.**

**D** is incorrect as this option does not include butanedoic acid which does react with calcium to produce a compound with the empirical formula $CaC_2H_3O_2$.

4

The correct answer is **A** because:

- Step 1: the **ester**, $C_{15}H_{31}CO_2C_{16}H_{33}$, is hydrolysed which requires either dilute acid or **dilute alkali** and heat:

$$C_{15}H_{31}CO_2C_{16}H_{33} \rightarrow C_{15}H_{31}CO_2H + C_{16}H_{33}OH$$

**ester → carboxylic acid + alcohol**

- Step 2: the carboxylic acid product, $C_{15}H_{31}CO_2H$, reacts with the sodium hydroxide in a neutralisation reaction to produce $C_{15}H_{31}CO_2Na$ and water:

$$C_{15}H_{31}CO_2H + NaOH \rightarrow C_{15}H_{31}CO_2Na + H_2O$$

**carboxylic acid + alkali → salt + water**

- Therefore, the total products are option **A** - $C_{15}H_{31}CO_2Na$ and $C_{16}H_{33}OH$.

**B** is incorrect as sodium does not replace hydrogen in alcohol groups, only in carboxylic acids.

**C** is incorrect as the structural formula $C_{15}H_{31}CO_2C_{16}H_{33}$ shows the carboxylic acid part of the ester is before the ester bond and the alcohol part is after, this option gets this the wrong way around.

**D** is incorrect as sodium does not replace hydrogen in alcohol groups, only in carboxylic acids. The structural formula $C_{15}H_{31}CO_2C_{16}H_{33}$ shows the carboxylic acid part of the ester is before the ester bond and the alcohol part is after, this option gets this the wrong way around

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**carboxylic acid** + **alkali** → **salt** + **water**

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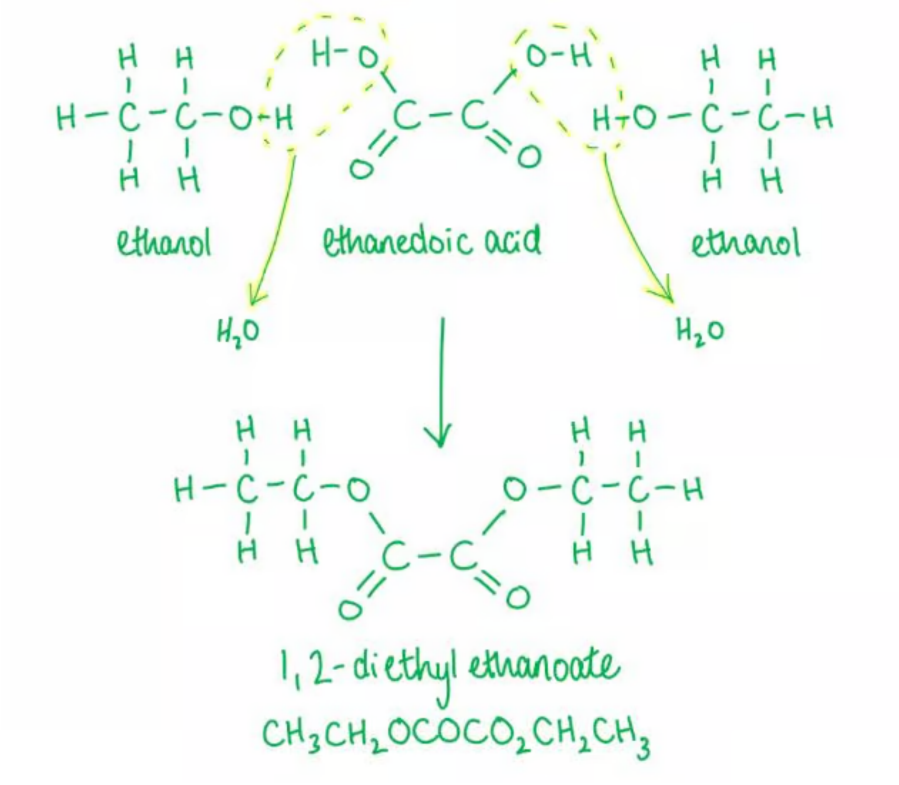
5

The correct answer is **B** because:




6

The correct answer is **D** because:

- A 'hydroxyl' group is an '-OH' group which in this molecule can be seen in both carboxylic acid groups and the alcohol group.
- If the substance only reacts with one hydroxyl group this must be the **alcohol**, therefore we must look at what **doesn't react** with a carboxylic acid group.
- The **dichromate ion**, $Cr_2O_7^{2-}$, which is produced when the ionic potassium dichromate, $K_2Cr_2O_7$, is dissolved in **aqueous solution** and **acidified**, $H^+$, acts as an **oxidising agent.**
- Carboxylic acid groups are oxidised as far as possible and therefore will **not** react with acidified dichromate ions.
- **Alcohols** can be **oxidised** by acidified dichromate ions:
    - First to a **ketone** in this case (secondary alcohols) or to an **aldehyde** (primary alcohols).
    - Ketones are **not** oxidised further.
    - However, aldehydes can be **further oxidised** to a **carboxylic acid.**

**A** is incorrect as **sodium** metal reacts with both **carboxylic acid** groups and alcohol group in **redox** reactions to produce a **salt** and **hydrogen** gas.
$2RCOOH + 2Na \rightarrow 2RCOONa + H_2$ & $2RCOH + 2Na \rightarrow 2RCONa + H_2$.

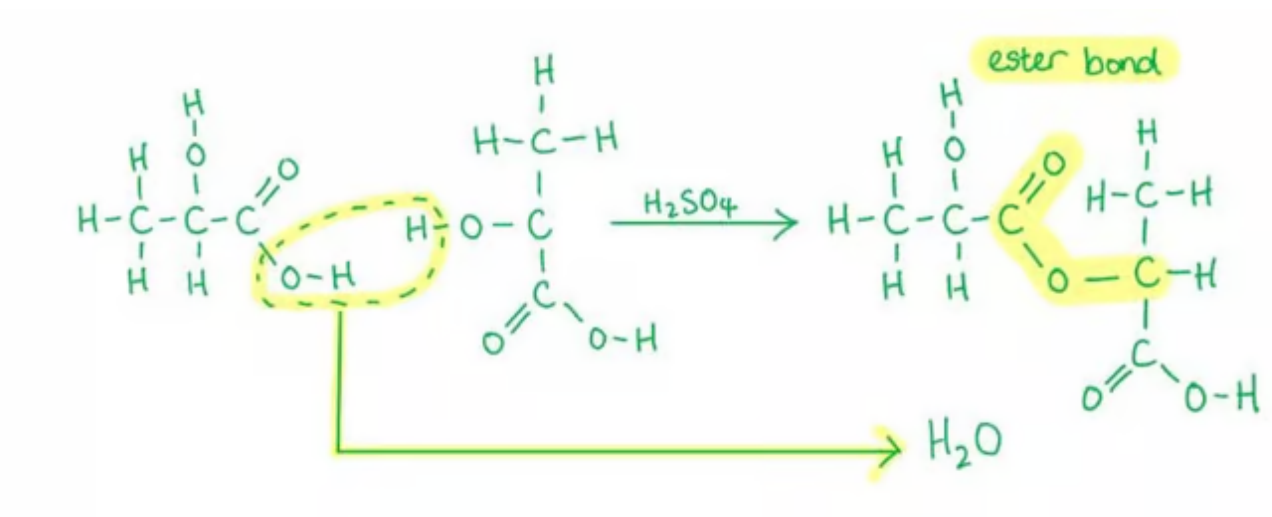
**B** is incorrect as $PCl_5$ reacts with both **carboxylic acid** groups to produce an acid chloride:
$RCOOH + PCl_5 \rightarrow RCOOCl + POCl_3 + HCl$.
$PCl_5$ reacts with **alcohols** to produce a **haloalkane** in a **nucleophilic substitution** reaction:
$RCH_2OH + PCl_5 \rightarrow RCH_2Cl + POCl_3 + HCl$.

**C** is incorrect as **sodium hydroxide** will react with both carboxylic acid groups to produce a salt and water in the follow neutralisation reaction:
$RCOOH + NaOH \rightarrow RCOONa + H_2O$.

7

The correct answer is **B** because:

- Lactic acid contains both an alcohol and a carboxylic acid functional group.
- In the presence of **sulfuric acid**, which acts as a **catalyst**, alcohol and carboxylic acid groups undergo a **condensation** reaction to produce an **ester** and water.



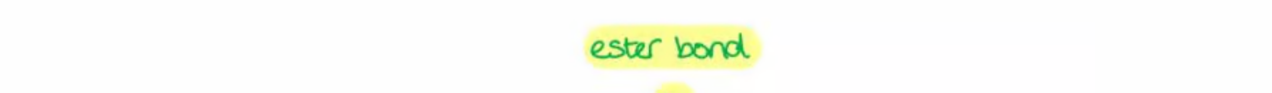

**A** is incorrect as aqueous bromine is rapidly decolourised by alkenes when the electrophilic addition of $Br_2$ occurs to form a halogenoalkane.

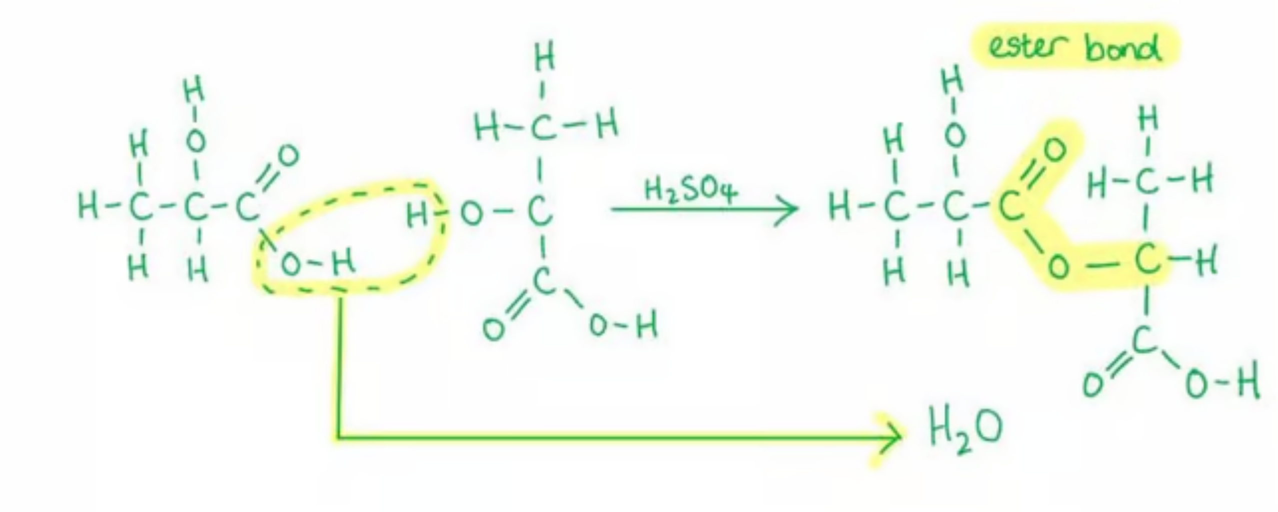
**C** is incorrect as Fehling's solution is only reduced by aldehydes.

**D** is incorrect as lactic acid is soluble in water because the alcohol and carboxylic acid groups form hydrogen bonds with water molecules.

8

The correct answer is **C** because:



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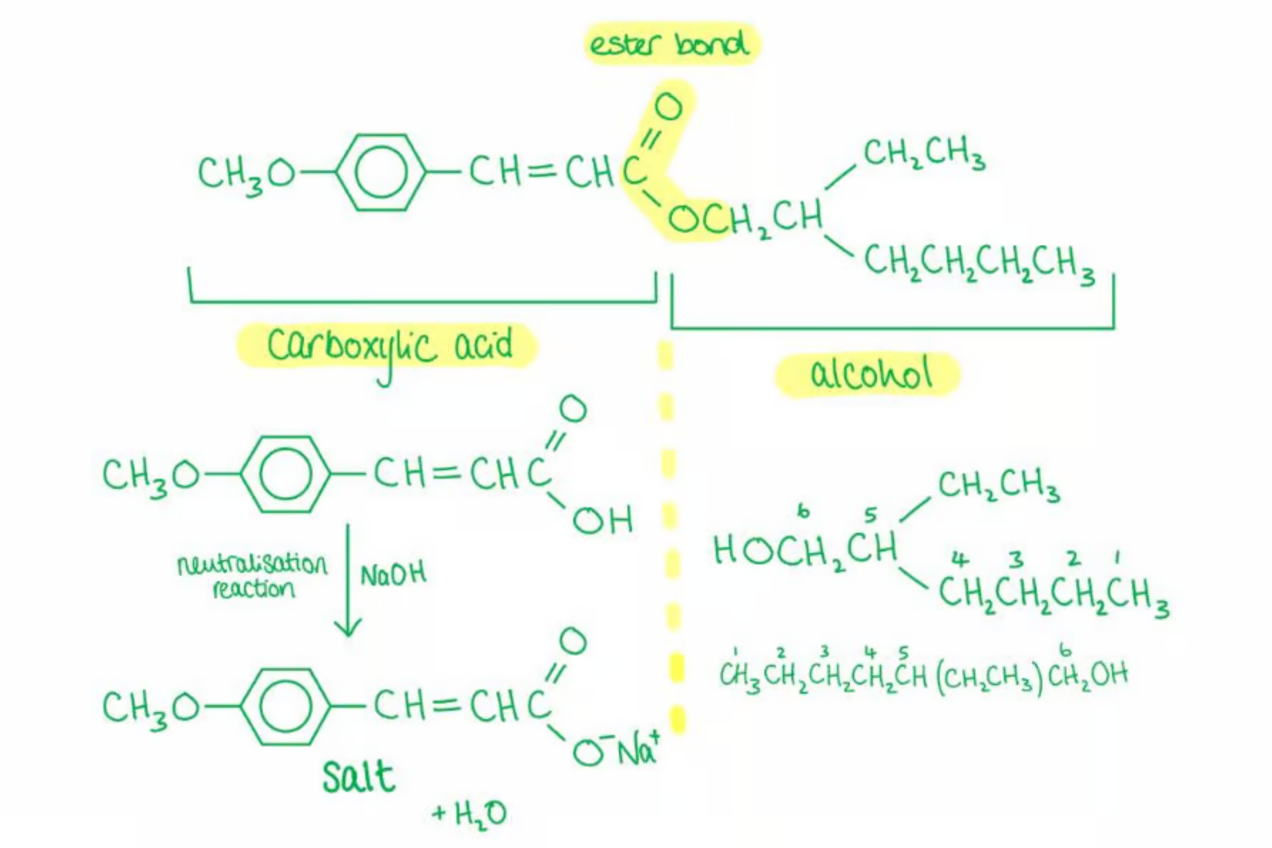


**A, B** & **D** are incorrect as $CO_2^-Na^+$ would not be produced from the alkene functional group in compound 1, only from the carboxylic acid group formed when the ester bond is broken – this is further along the compound.

9

The correct answer is **A** because:

- Compound Q is an **ester** made from propanoic acid, $CH_3CH_2CO_2H$, and methanol, $CH_3OH$.
- The naming of the ester starts with the alkyl attached to the alcohol – in this case methyl – and ends with the carboxylic acid, in this case propanoate.
  - '-oate' is the suffix of compounds produced from carboxylic acids.
- Therefore, the name is **methyl propanoate.**
- The boiling point of methyl propanoate is **lower** than butanoic acid because it has only **dipole-dipole attractions** between the molecules, whereas butanoic acid has stronger **hydrogen bonds.**

10

The correct answer is **C** because: